

MetaFVG — Spearman Mean-Reversion Gate

Equal-Weight Portfolio Report — 14 Instruments — M15/4h — Fixed 2% Notional Sizing

Strategy Recap

MetaFVG trades Fair Value Gaps detected on a higher timeframe (HTF, 4h resample) and confirmed with lower-timeframe (LTF, M15) momentum before entry. For each HTF FVG zone, price re-entering the gap with a Fibonacci-style fill percentage triggers a search for a strong LTF momentum candle pattern in the trigger direction; on confirmation, a trade is opened with a pivot-based stop loss and an ATR-scaled take profit.

This report covers the **mean-reversion variant**: a rolling Spearman rank-correlation gate is computed on HTF closes, and whenever the gate's sign *disagrees* with the HTF zone's own direction (`invert_direction=True`), the trade is taken in the gate's direction instead of the zone's — both the LTF momentum confirmation and the trade parameters follow the disagreeing signal, not the original zone. Gate threshold: rolling window 20 bars, $|p|$ gate acceptance as calibrated in the FX Majors sweep.

The 14-instrument universe (AUDUSD, EURUSD, GBPUSD, NZDUSD, USDCAD, USDCHF, USDJPY, US500, GER40, JP225, HK50, USDSGD, EURNOK, USDZAR) was deliberately assembled to be fundamentally diversified across central banks and macro drivers, not just statistically decorrelated — four equity indices from different central banks plus three FX minors add currency exposures with no recombination of the 7 FX Majors. This report blends all 14 with **equal weights (1/14 each)**; a companion study tested four non-equal-weight allocation schemes and found none improved on equal weighting's Sharpe (see [weighting_schemes/comparison_summary.pdf](#)).

Process Diagram

HTF FVG detection → LTF momentum confirmation → regression/Spearman/Lasso gate → trade parameter calculation → pivot SL / ATR TP.

MetaFVG + Regression Gate

Signal generation & risk management pipeline -- HTF=4h / LTF=M15 -- reference setup

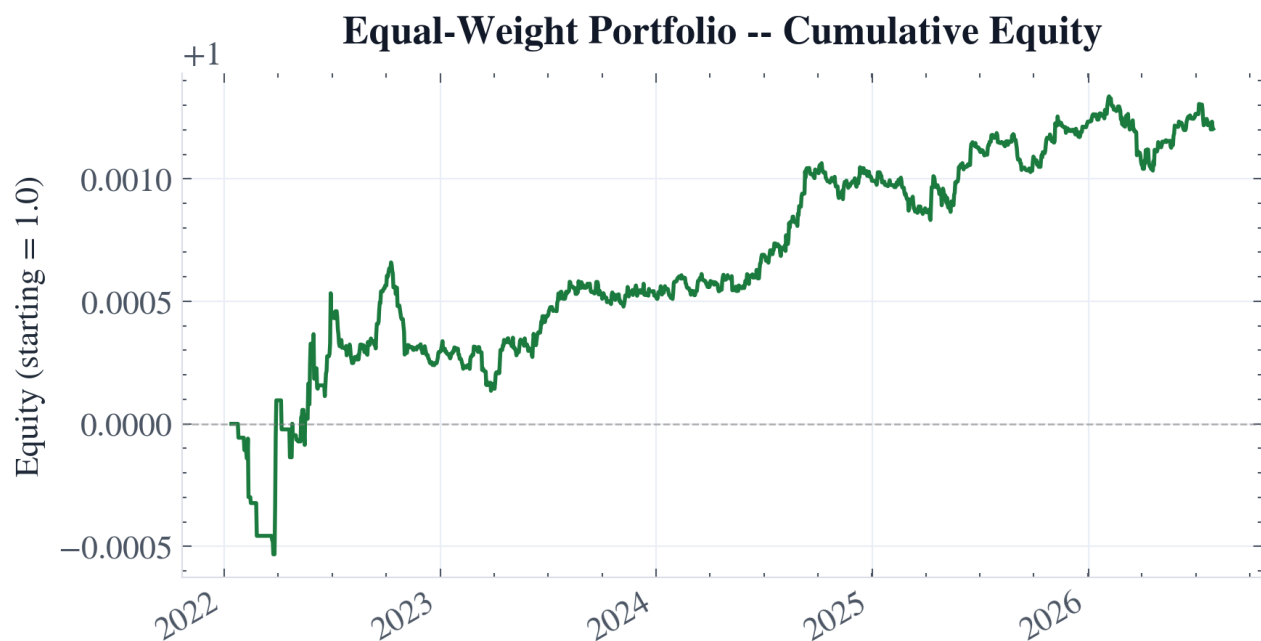


Post lookahead-bias-fix reference setup: avg individual Sharpe +0.010, blended 7-pair FX Majors portfolio Sharpe +0.143

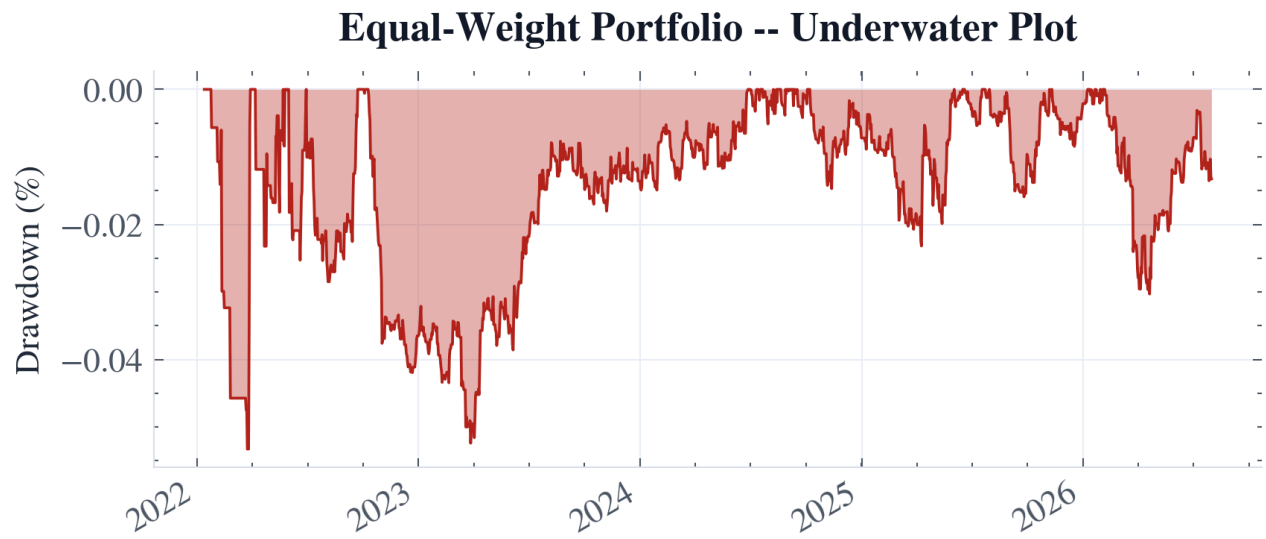
Portfolio Results

Metric	Value
Total Return	+0.120%
CAGR	+0.0183%
Sharpe (daily, ann.)	0.518
Sortino	0.817
Calmar	0.343
Max Drawdown	-0.0533%
Annualized Volatility	0.0353%
Win Rate (days)	30.89%
Best Day	+0.0291%
Worst Day	-0.0239%
Skew	1.545
Tail Ratio	1.239
Kelly Criterion	0.0668
VaR (95%)	-0.0036%
CVaR (95%)	-0.0072%
Recovery Factor	2.260

Equity Curve



Underwater Plot



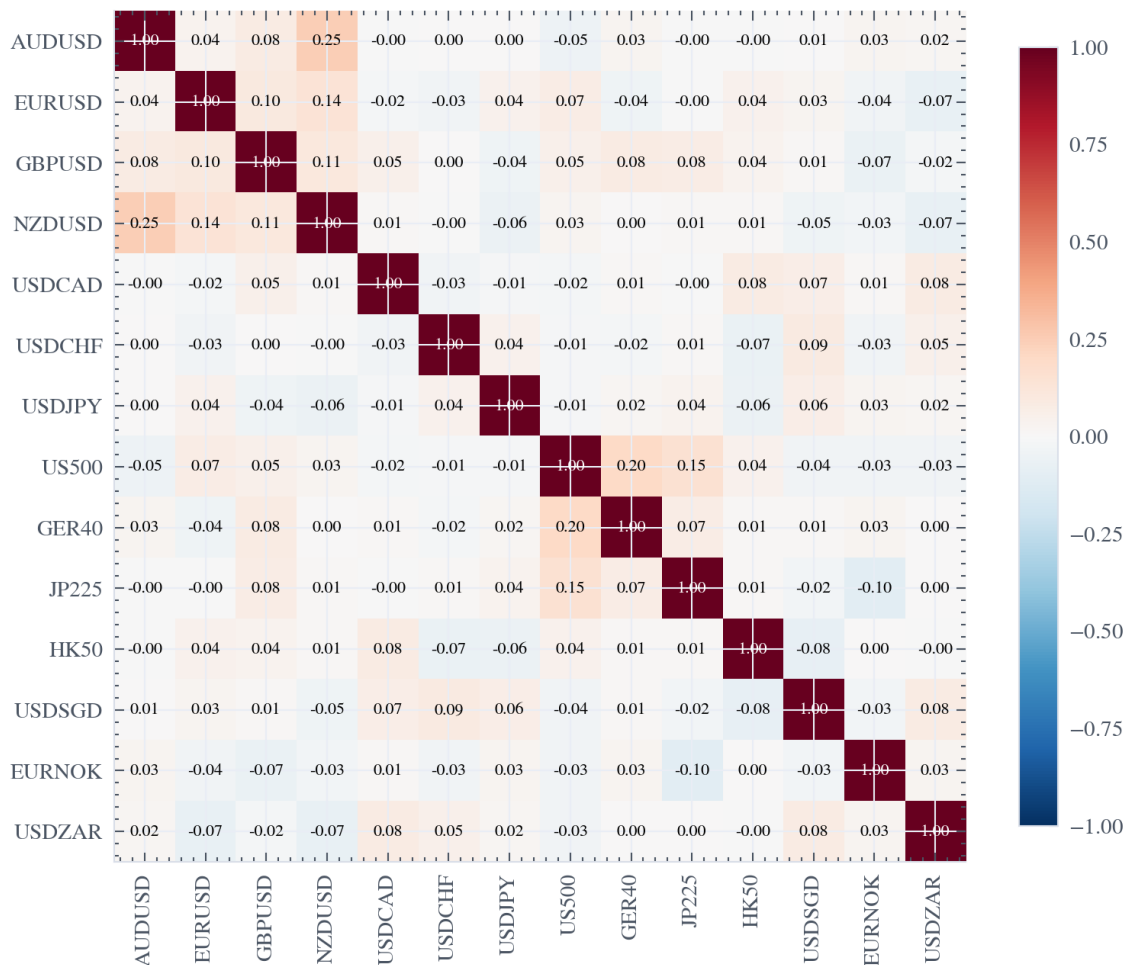
Rolling Correlation Between Assets

Average pairwise Pearson correlation across the 14 instruments' daily strategy P&L, computed on a rolling 180-calendar-day window stepped every 14 days (minimum 15 overlapping observations per pair). This measures realized co-movement of the *traded* P&L streams, not the underlying instruments' raw price returns — the latter are structurally far more correlated (e.g. the 7 FX Majors alone run 0.3–0.9 in absolute price-return correlation via a common USD factor); sparse, largely non-overlapping FVG trade timing across instruments is what keeps the traded correlation this low, on top of the fundamental diversification built into instrument selection.

Rolling 180-Day Average Pairwise Correlation (14 instruments)



Full-Sample Pairwise Correlation Matrix



Per-Symbol Performance

Standalone performance of each instrument's own trade stream (fixed 2% notional sizing), sorted by Sharpe. Equal weighting means each instrument contributes 1/14 of portfolio capital regardless of its standalone Sharpe.

Symbol	Closed Trades	Win Rate	Sharpe	Max DD	Total Return
NZDUSD	352	28.7%	1.141	9.86%	+0.333%
GBPUSD	331	31.7%	1.092	19.42%	+0.213%
HK50	347	28.2%	0.958	20.31%	+0.588%
US500	399	28.6%	0.806	15.94%	+0.350%
AUDUSD	345	24.9%	0.799	12.84%	+0.250%
JP225	362	29.3%	0.541	29.92%	+0.369%
USDCHF	339	25.7%	0.476	11.22%	+0.099%
USDJPY	340	25.6%	-0.130	16.43%	-0.031%
EURNOK	241	22.0%	-0.305	17.07%	-0.077%
GER40	384	24.2%	-0.476	31.66%	-0.195%
USDCAD	347	22.8%	-0.477	15.03%	-0.084%
USDZAR	359	20.6%	-0.609	38.70%	-0.221%
EURUSD	374	20.9%	-0.634	17.04%	-0.111%
USDSGD	319	18.5%	-0.691	16.89%	-0.102%